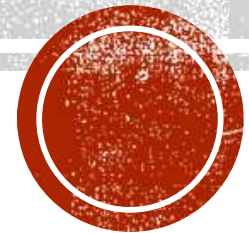




“MODULANDIA” GEOCITY – THE CREATION OF SYMMETRY.

**Vilnius Žemyna Gymnasium
Project Supervisor : Sigita Kaminskienė
LITHUANIA**



Work objective – to explore symmetry and create a futuristic city based on ornaments.

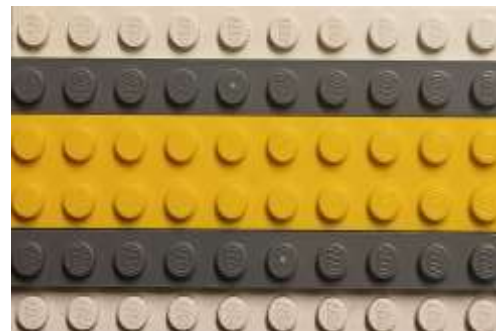
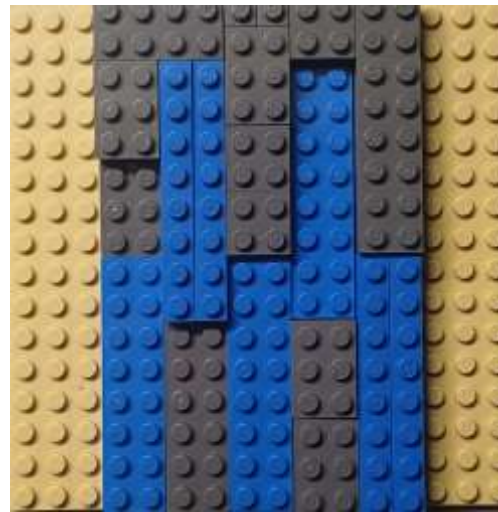
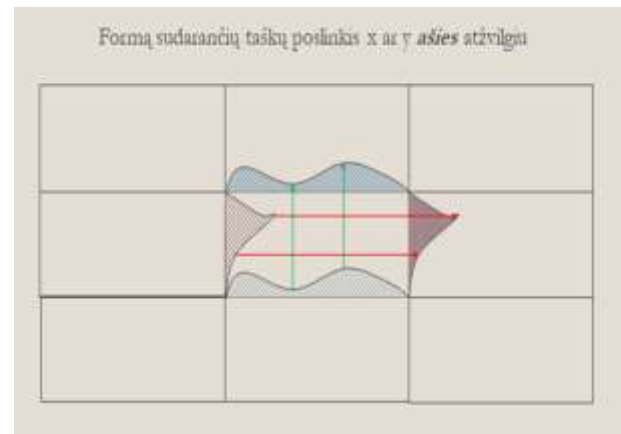
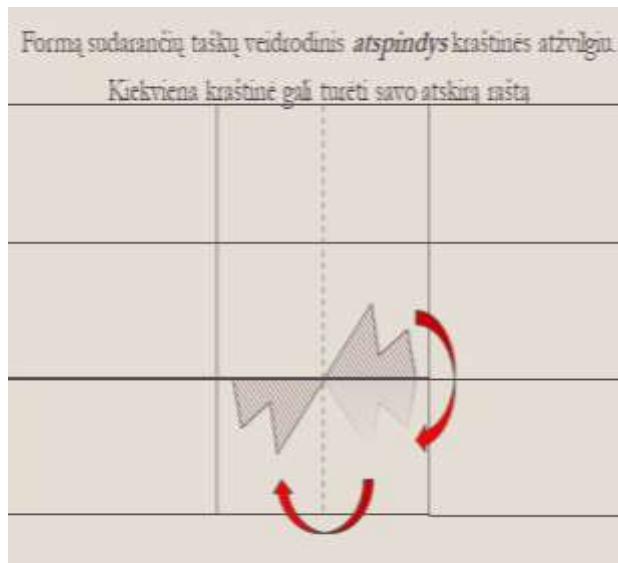
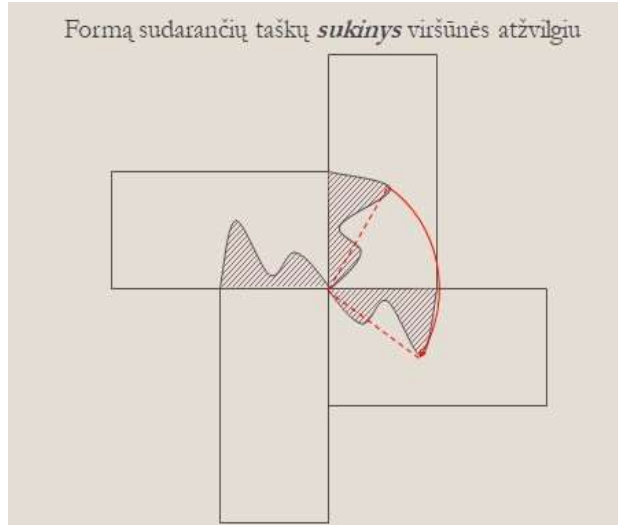
Objectives:

- To identify the types of symmetry: axial, angular, translational, rotational, and reflective symmetry.**
- To create original ornaments by applying harmony of colors and shapes.**
- To use digital software for ornament design and city visualizations.**
- To design ornaments using 3D modules.**
- To create a visual layout of the city plan.**
- To develop creativity, critical thinking, and technological skills.**



The process / stages
of creating
tessellating shapes.

Using:
axial translational
symmetry;
rotational
symmetry;
reflective
symmetry.



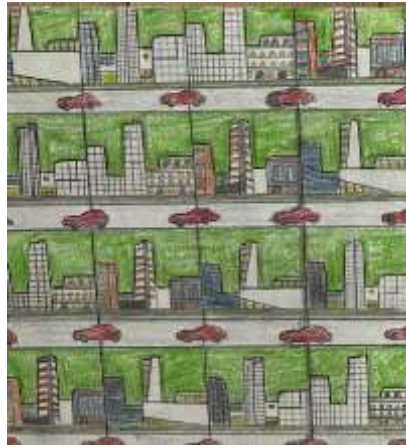
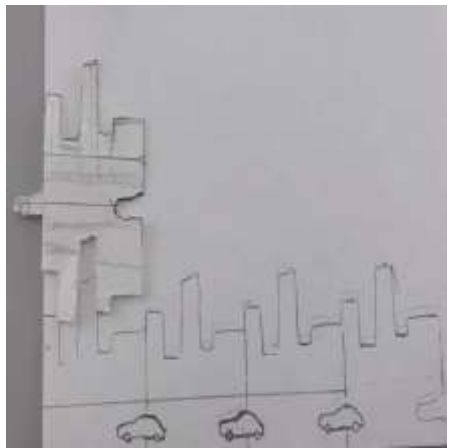
VECTOR GRAPHICS. AXIAL TRANSLATIONAL SYMMETRY.



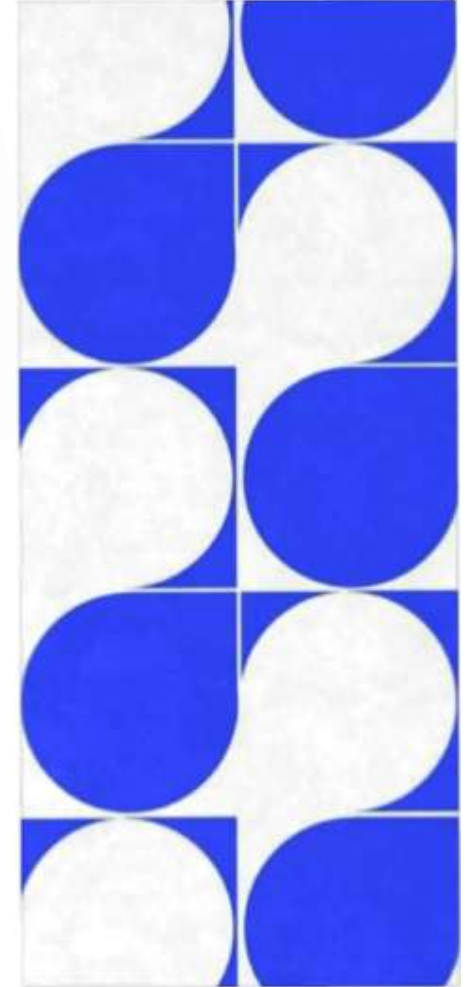
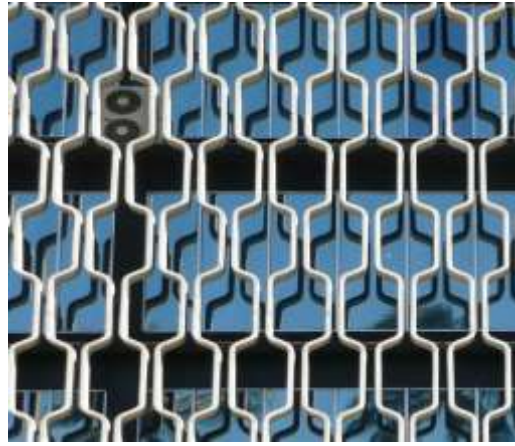
PROCESS OF
CREATING
TESSELLATING
Gt (SURFACE
-COVERING)
SHAPES.



Students' work results:



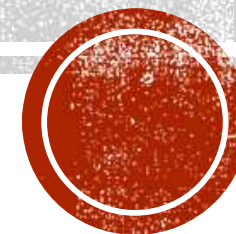
Where it is found and applied in real life:





**„MODULANDIA.“ THE CITY OF THE FUTURE: A SUSTAINABLE
LIVIG ENVIRONMENT“.**

MATHEMATICS – INFORMATICS – ARTS- GEOGRAPHY.

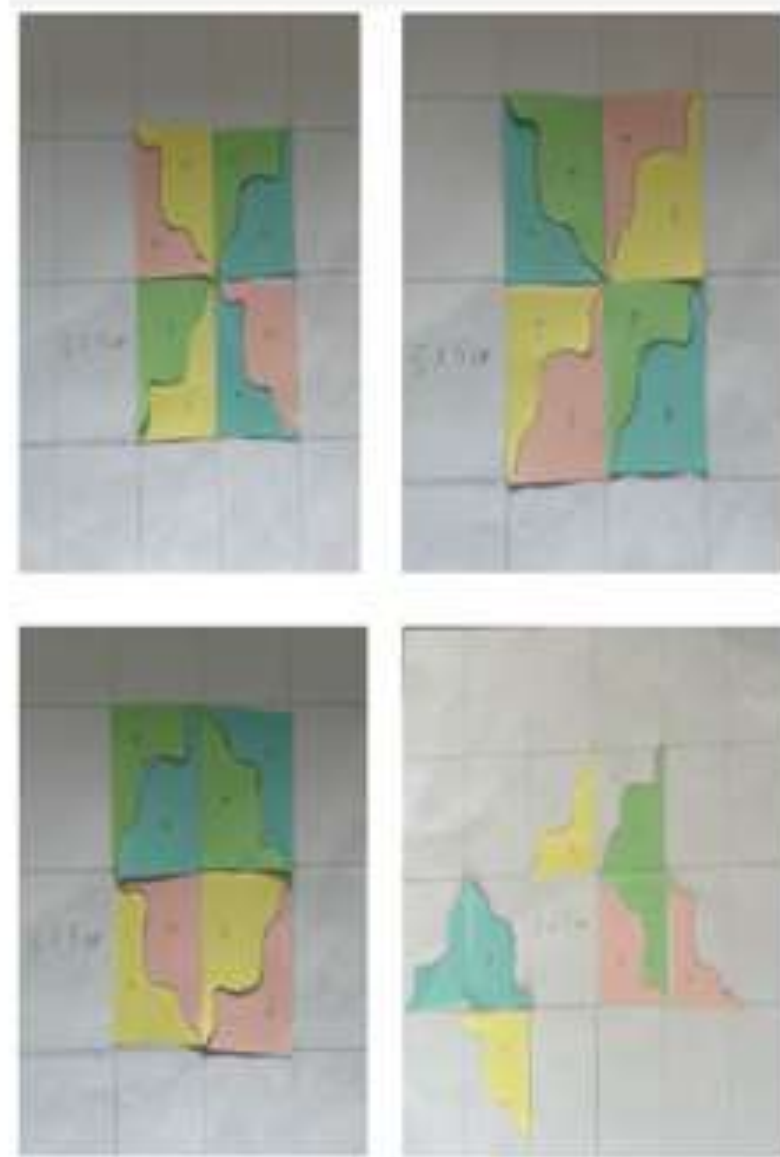


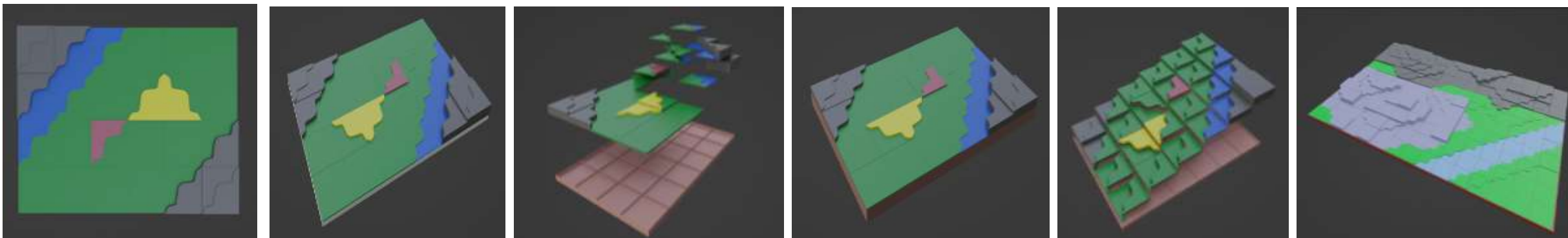
Objective:

Using a mathematical formula for tessellating ornaments, and by creating a combination of ornaments that can be rotated or tilted in any direction, we obtain a constantly changing visual pattern.

By 3D-printing buildings onto these ornamental structures using a 3D printer, we create a city that changes its layout and appearance.

Everyone can design their own city according to their needs.

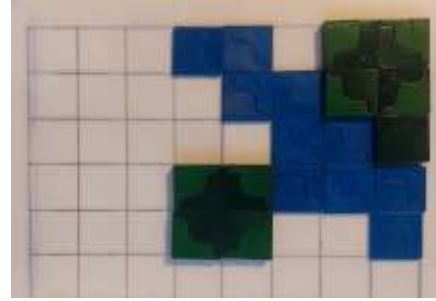
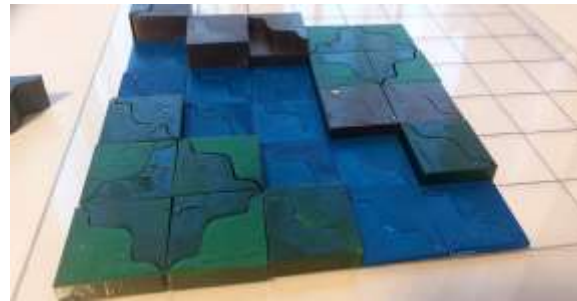
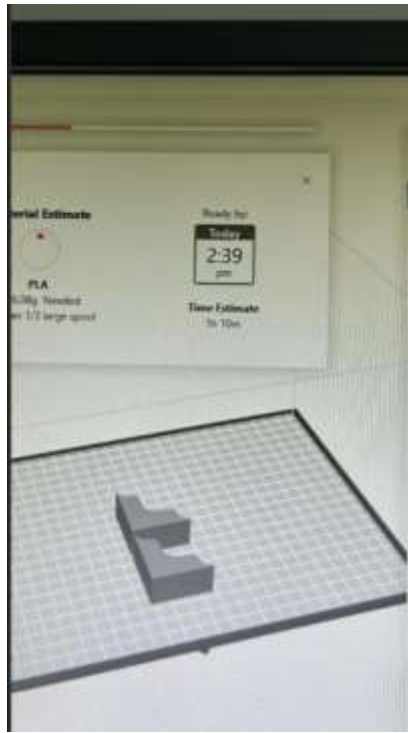




Blender visualizes

The Modulandia city project was modeled using the Blender software, creating a product prototype that can be practically adapted by anyone. This allows for multiple interpretations and enables faster visualization of the desired outcome. Laser cutting, a 3D printer, and technical drawings were used in the development process. This project is suitable for addressing topics in environmental engineering. Related AI theme: “The city of the future: a sustainable living environment.”





RESULTS:

- An innovative Modulandia city model was successfully created, which is functional and practical. We are satisfied with the results; the work is unique, and we improved our 3D modeling skills. Such creative projects allow us to realize innovative ideas by combining science and art.





During the lesson, we revised and learned:

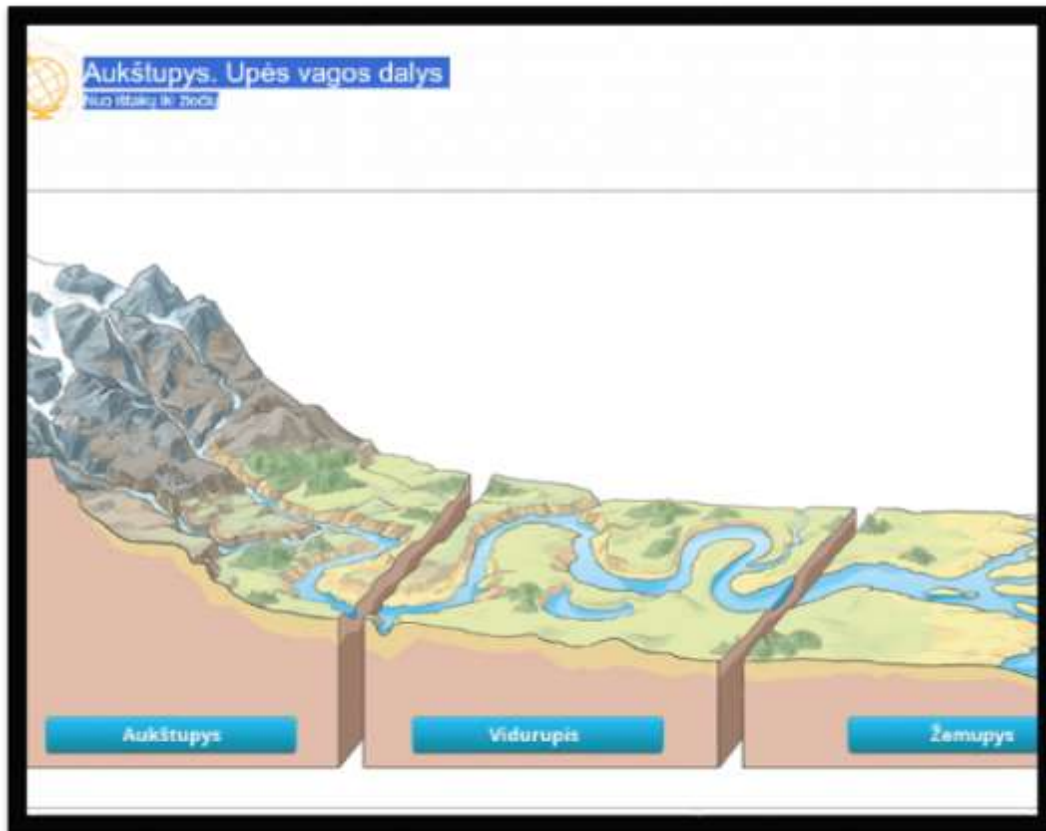
- what shape the Earth is;
- how it was imagined by Pythagoras and Aristotle;
- what geodesy is;
- how GPS works;
- about the profession of a surveyor.

We also learned about the principles of laser scanning and tested the VRScan3D simulator.

We studied how scanning works, simulated data collection, created 3D point clouds, and used virtual environments such as CloudCompare.



IMPLEMENTATION OF THE IDEA IN GEOGRAPHY.



- The acquired knowledge and applied tools are used to analyze sections of the Upper River channel, assessing their formation, characteristics, and significance for the landscape. This model allows not only the creation of forms but also the analysis of geographical processes such as relief, water movement, and urban development. By modifying the modules, different territorial scenarios can be modeled and evaluated to determine which solution is more sustainable.



CONCLUSIONS

The project was successfully implemented – a functional and practical modular city model was created, allowing easy interpretation and visualization of different solutions. The work demonstrated that 3D modeling combined with physical model creation is an effective method for analyzing environmental engineering processes. These skills open career opportunities in design, geodesy, architecture, and industry, and can also be applied in computer game development.

SUMMARY

The project combined digital modeling and practical fabrication using various technologies. During the process, 3D modeling skills were improved, creative abilities were expanded, and a better understanding of landscape formation principles was achieved. The created model enables flexible adaptation and analysis of different urban development scenarios.

RECOMMENDATIONS

It is recommended to further improve the precision and detail of the model to achieve greater realism. It is also worth expanding its application in education and research, integrating additional analysis tools and interactive functions.

